

(Question 1)

(a)

Since we are evaluating function, x in the function will be our data d .

Therefore, the change of data $\delta d = (x + h) - x$

On the other hand, the output of our data x , will be $f(x)$

After data perturbed, δx will be $f(x + h) - f(x)$

From eqn. 2.4, $K(d) = \sup_{\delta d \in D} \frac{\|\delta x\|}{\|x\|} / \frac{\|\delta d\|}{\|d\|}$

Since $f(x) : \mathbb{R} \rightarrow \mathbb{R}$, $\|x\| = |x|$, $\|f(x)\| = |f(x)|$

$$\begin{aligned} & \sup_{\delta d \in D} \frac{|\delta x|}{|x|} / \frac{|\delta d|}{|d|} \\ &= \sup_{h \in D} \frac{|f(x+h) - f(x)|}{|f(x)|} \cdot \frac{|x|}{|(x+h) - x|} \\ &= \lim_{h \rightarrow 0} \sup_{h \in D} \frac{|f(x+h) - f(x)|}{|h|} \cdot \frac{|x|}{|f(x)|} \\ &= |f'(x)| \cdot \frac{|x|}{|f(x)|} \end{aligned}$$

(b)

$$\begin{aligned} K_{abs}(d) &= \sup_{\delta d \in D} \frac{\|\delta x\|}{\|\delta d\|} \\ &= \sup_{\delta d \in D} \frac{|\delta x|}{|\delta d|} \\ &= \sup_{h \in D} \frac{|f(x+h) - f(x)|}{|(x+h) - x|} \\ &= \lim_{h \rightarrow 0} \sup_{h \in D} \frac{|f(x+h) - f(x)|}{|h|} \\ &= |f'(x)| \end{aligned}$$

(c)

$$\begin{aligned} f(x) &= \frac{\beta + x}{\beta - x} \\ f'(x) &= \frac{\frac{d}{dx}(\beta + x) \cdot (\beta - x) - \frac{d}{dx}(\beta - x) \cdot (\beta + x)}{(\beta - x)^2} \\ &= \frac{(\beta - x) - (1 - (\beta + x))}{(\beta - x)^2} \\ &= \frac{\beta - x + \beta + x}{(\beta - x)^2} \\ &= \frac{2\beta}{(\beta - x)^2} \\ K_{rel}(x) &= \left| x \frac{2\beta}{(\beta - x)^2} / \frac{\beta + x}{\beta - x} \right| \\ &= \left| x \frac{2\beta}{(\beta - x)^2} \cdot \frac{\beta - x}{\beta + x} \right| \\ &= \left| \frac{2\beta x}{\beta^2 - x^2} \right| \\ K_{abs}(x) &= \left| \frac{2\beta}{(\beta - x)^2} \right| \end{aligned}$$

Compute condition numbers around $x = 1$ and $x = 100$ when $\beta = 1.01$

$K_{rel}(x)$ around 1:

```

x = 0.9, k_rel = 8.65302237029986
x = 0.95, k_rel = 16.318027210884345
x = 0.99, k_rel = 49.994999999999955
x = 1, k_rel = 100.49751243781091
x = 1.02, k_rel = 101.49753694581288
x = 1.05, k_rel = 25.740291262135912
x = 1.1, k_rel = 11.700895208004203

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$K_{abs}(x)$ around 1:

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x = 0.9, k_abs = 166.9421487603306
x = 0.95, k_abs = 561.1111111111101
x = 0.99, k_abs = 5049.999999999991
x = 1, k_abs = 20199.999999999964
x = 1.02, k_abs = 20199.999999999964
x = 1.05, k_abs = 1262.4999999999977
x = 1.1, k_abs = 249.38271604938225

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$K_{rel}(x)$ around 100:

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x = 0.9, k_rel = 8.65302237029986
x = 0.95, k_rel = 16.318027210884345
x = 0.99, k_rel = 49.994999999999955
x = 1, k_rel = 100.49751243781091
x = 1.02, k_rel = 101.49753694581288
x = 1.05, k_rel = 25.740291262135912
x = 1.1, k_rel = 11.700895208004203

```

$K_{abs}(x)$ around 100:

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x = 99.9, k_rel = 0.020222287227688242
x = 99.95, k_rel = 0.020212168959272567
x = 99.9, k_rel = 0.020222287227688242
x = 100, k_rel = 0.020202060812223456
x = 100.01, k_rel = 0.02020004039604119
x = 100.05, k_rel = 0.020191962771360254
x = 100.1, k_rel = 0.02018187482153266

```

- As x increases, K_{abs} decreases significantly.
- K_{rel} also decreases as x increases, but not as significantly as K_{abs}
- Generally, K changes significantly when x changes slightly.
- $K(x)$ is large when $x = 1$
- When $x = 1.01$, $K(x)$ undefined, which is ill-conditioned.
- When $x = 1$ or 1.02 (around 1.01), $K(x)$ will be a large number, which is ill-conditioned.
- As x getting further away from 1 , $K(x)$ will become smaller and smaller.
- When x is around 100 , $K(x)$ is always a small number.

Overall, $f(x)$ is ill-conditioned when x is very close to 1.01 . It's well-conditioned when x is away from 1.01 .

(Question 2)

a)

What happens to the roots as $\beta \rightarrow \infty$

Observe $p(x) = x^2 + 2\beta x + 1$, as $\beta \rightarrow \infty$, the dominant term of $p(x)$ will be $2\beta x$. Hence, x^2 will gradually act like a constant in the function.

So, as $\beta \rightarrow \infty$, $p(x) \approx 2\beta x + c$ which is nearly a linear function.

Setting $2\beta x + c = 0$ for finding root, $x = -\frac{c}{2\beta}$. As $\beta \rightarrow \infty$, the root will be $\rightarrow 0_-$

On the other hand, since it's only approximately close to a linear function, we would still need to consider the other root.

The second root of the function $-\beta - \sqrt{\beta^2 - 1}$ is $\approx -\beta - \beta$ as $\beta \rightarrow \infty$. Which implies $-2\beta \approx -\infty$. In this case the second root will be $= \infty$.

b)

Since G is a vector-valued function, use $\|v\|_2 = \sqrt{v_1^2 + v_2^2} = x_+(\beta) = v_1, x_-(\beta) = v_2$

$$\begin{aligned} x_+ : -\beta + \sqrt{\beta^2 - 1}, \frac{d}{d\beta} &= -\beta + (\beta^2 - 1)^{\frac{1}{2}} = \frac{1}{2} (\beta^2 - 1)^{-\frac{1}{2}} \cdot 2\beta = -1 + \frac{\beta}{\sqrt{\beta^2 - 1}} \\ x_- : -\beta - \sqrt{\beta^2 - 1}, \frac{d}{d\beta} &= -\beta - (\beta^2 - 1)^{\frac{1}{2}} = -\frac{1}{2} (\beta^2 - 1)^{-\frac{1}{2}} \cdot 2\beta = -1 - \frac{\beta}{\sqrt{\beta^2 - 1}} \end{aligned}$$

From 2.19, If G is differentiable,

$$K_{rel}(\beta) = \|G'(\beta)\| \cdot \frac{\|\beta\|}{\|G(\beta)\|} \quad K_{abs}(\beta) = \|G'(\beta)\|$$

$$\begin{aligned} \|G'(\beta)\| &= \sqrt{(x'_+(\beta))^2 + (x'_-(\beta))^2} \\ &= \sqrt{\left(-1 + \frac{\beta}{\sqrt{\beta^2 - 1}}\right)^2 + \left(-1 - \frac{\beta}{\sqrt{\beta^2 - 1}}\right)^2} \\ &= \sqrt{2(1) + 2\left(\frac{\beta}{\sqrt{\beta^2 - 1}}\right)^2} \\ &= \sqrt{2 + 2\frac{\beta^2}{\beta^2 - 1}} \\ \|G(\beta)\| &= \sqrt{(-\beta + \sqrt{\beta^2 - 1})^2 + (-\beta - \sqrt{\beta^2 - 1})^2} \\ &= \sqrt{2(\beta)^2 + 2(\sqrt{\beta^2 - 1})^2} \\ &= \sqrt{2\beta^2 + 2(\beta^2 - 1)} \\ &= \sqrt{4\beta^2 - 2} \\ K_{rel}(\beta) &= \sqrt{2 + 2\frac{\beta^2}{\beta^2 - 1}} \cdot \frac{|\beta|}{\sqrt{4\beta^2 - 2}} \\ &= |\beta| \cdot \sqrt{\frac{2 + 2\frac{\beta^2}{\beta^2 - 1}}{4\beta^2 - 2}} \\ &= |\beta| \cdot \frac{\left(1 + \frac{\beta^2}{\beta^2 - 1}\right)}{(2\beta^2 - 1)} \\ &= |\beta| \cdot \frac{2\beta^2 - 1}{\beta^2 - 1} \cdot \frac{1}{2\beta^2 + 1} \\ &= |\beta| \cdot \frac{1}{\beta^2 - 1} \\ K_{rel}(\beta) &= \frac{|\beta|}{\sqrt{\beta^2 - 1}} \quad K_{abs}(\beta) = \sqrt{\frac{2\beta^2 - 1}{\beta^2 - 1}} \end{aligned}$$

c)

When β is away from 1, $K_{rel}(1.00001) = 223.6085$ to $K_{rel}(1.01) = 7.1240$. $K_{rel}(\beta)$ is getting smaller as β gets farther away from 1. Which makes the problem more wellconditioned.

As $\beta \rightarrow \infty$, $K_{\text{rel}}(\beta)$ will converge to 1 since $K_{\text{rel}}(B)$ is Small, $\beta \rightarrow \infty$ will make $G(\beta)$ well conditioned.

G will be ill-conditioned if β is close to 1 since $K_{\text{rel}}(1.01) = 7.1240$ and $K_{\text{rel}}(1.00001) = 223.6085$. $K_{\text{rel}}(\beta)$ is getting larger as β gets close to 1 which makes G ill-conditioned.

(Question 3)

Exact value = 5.015

Evaluate algorithm with rounding

Let $x = 5.01$ & $y = 5.02$

$$\alpha_1 = \frac{(5.01+5.02)}{2.0} = \frac{10.0}{2.0} = 5.00$$

$$\alpha_2 = \left(\frac{5.01}{2.0}\right) + \left(\frac{5.02}{2.0}\right) = 2.51 + 2.51 = 5.02$$

$$\alpha_3 = 5.01 + \left(\frac{(5.02-5.01)}{2.0}\right) = 5.01 + \left(\frac{0.01}{2.0}\right) = 5.01 + 0.005 = 5.015$$

Evaluate algorithm with chopping.

$x = 5.01, y = 5.02$

$$\alpha_1 = \frac{(5.01+5.02)}{2.0} = \frac{10.0}{2.0} = 5.00$$

$$\alpha_2 = \left(\frac{5.01}{2.0}\right) + \left(\frac{5.02}{2.0}\right) = 2.50 + 2.51 = 5.01$$

$$\alpha_3 = 5.01 + \left(\frac{(5.02-5.01)}{2.0}\right) = 5.01 + \left(\frac{0.01}{2.0}\right) = 5.01 + 0 = 5.01$$

When we evaluate the algorithm using rounding,

$$\varepsilon_{\text{rel}}(\alpha_1) = \frac{|5.015 - 5.00|}{|5.015|} = 0.003$$

$$\varepsilon_{\text{rel}}(\alpha_2) = \frac{|5.015 - 5.02|}{|5.0151|} = 9.970 \times 10^{-4} = 0.001$$

$$\varepsilon_{\text{rel}}(\alpha_3) = 0.001$$

α_1 has a higher relative error than other is because it rounded off $5.01 + 5.02$ in the beginning. Rounding 10.03 to 10.0, after divided by 2.0 will become 5.0 which is farther away from the true value.

α_2, α_3 both gave the correct result but α_3 is able to get the true value but was rounded to 5.02 because of the value of t .

After the first step of α_2 , if it didn't round off 2.505 to 2.51, α_2 will be able to get the true value too.

When we evaluate the algorithm using chopping,

$$\varepsilon_{\text{rel}}(\alpha_1) = \frac{|5.015 - 5.00|}{|5.015|} = 0.003$$

$$\varepsilon_{\text{rel}}(\alpha_2) = \frac{|5.015 - 5.01|}{|5.015|} = 0.001$$

$$\varepsilon_{\text{rel}}(\alpha_3) = 0.001$$

α_1 has a higher relative error than other is because it chopped off 5.01 + 5.02 in the beginning. Chopping 10.03 to 10.0, after divided by 2.0 will become 5.0 which is farther away from the true value.

α_2, α_3 both gave the correct result but α_3 is able to get the true value but was chopped to 5.02 because of the value of t .

(Question 4)

Properties of Norm: P.18

- (1) Positivity $\|x\| \geq 0, \forall x \in X$ and $\|x\| = 0$ iff. $x = 0$
- (2) Homogeneity $\|cx\| = |c|\|x\|$
- (3) Triangular inequality $\|x + y\| \leq \|x\| + \|y\|$

Euclidean Norm (L2 Norm):

$$\|x\|_2 = \sqrt{\sum_{i=1}^n |x_i|^2} = (x_1^2 + x_2^2 + \dots + x_n^2)^{\frac{1}{2}}$$

Proof condition 1:

$\|\cdot\|_2$ will always satisfy positivity because sum of squares will always ≥ 0

Now, to prove iff. condition, $\|\vec{0}\|_2 = \sqrt{\sum_{i=1}^n |0|^2} = (0_1^2 + 0_2^2 + \dots + 0_n^2)^{\frac{1}{2}} = 0$

n -column 0 vector, to compute it's L_2 norm: $\sqrt{0_1^2 + 0_2^2 + \dots + 0_n^2} = \sqrt{\sum_{i=1}^n 0^2} = \|\vec{0}\|_2$

Proof condition 2:

$$\|cx\|_2 = \sqrt{\sum_{i=1}^n (cx_i)^2} = \sqrt{c^2 \sum_{i=1}^n x_i^2} = |c|\|x\|_2$$

Proof condition 3:

$$\begin{aligned} & \|x + y\|_2 \\ &= \|x + y\|_2^2 \text{ we can do this because } x, y \text{ is positive} \\ &= \left(\left(\sum_{i=1}^n (x_i + y_i)^2 \right)^{\frac{1}{2}} \right)^2 \\ &= \sum_{i=1}^n (x_i + y_i)^2 \\ &= \sum_{i=1}^n x_i^2 + y_i^2 + 2x_i y_i \\ &= \|x\|_2^2 + \|y\|_2^2 + 2 \cdot x \cdot y, \text{ where } x \cdot y \text{ is inner product} \\ & \text{From Cauchy: } x \cdot y \leq \|x\|_2 \|y\|_2 \end{aligned}$$

We can conclude:

$$\begin{aligned}
&\|x\|_2 + \|y\|_2 + 2x \cdot y \leq \|x\|_2 + \|y\|_2 + 2\|x\|_2\|y\|_2 \\
&(\|x + y\|_2)^2 \leq (\|x\|_2 + \|y\|_2)^2 \\
&\sqrt{(\|x + y\|_2)^2} \leq \sqrt{(\|x\|_2 + \|y\|_2)^2} \\
&\|x + y\|_2 \leq \|x\|_2 + \|y\|_2
\end{aligned}$$